

List 1- Finite Elements

1. Check the physical units of all quantities listed in Table 1.2.

Table 1.2: Physical interpretation of quantities in Poisson's problem. Units in SI.

Physical problem	u	F	$\vec{f}$	μ
Heat conduction	Temperature [$^{\circ}\text{K}$]	Heat source [$\frac{\text{W}}{\text{m}^3}$]	Heat flux [$\frac{\text{W}}{\text{m}^2}$]	Conductivity [$\frac{\text{W}}{\text{m}^{\circ}\text{K}}$]
Mass diffusion	Mass [$\frac{\text{mol}}{\text{m}^3}$]	Mass source [$\frac{\text{mol}}{\text{m}^3\text{s}}$]	Mass flux [$\frac{\text{mol}}{\text{m}^2\text{s}}$]	Diffusivity [$\frac{\text{m}^2}{\text{s}}$]
Flow in ducts	Velocity [$\frac{\text{m}}{\text{s}}$]	Pressure gradient [$\frac{\text{N}}{\text{m}^3}$]	Shear stress [Pa]	Viscosity [$\text{Pa} \cdot \text{s}$]

$$\nabla \cdot (\mu(x) \cdot \nabla u(x)) = F(x)$$

$$u: \Omega \subset \mathbb{R}^d \rightarrow \mathbb{R}$$

$$F, \mu: \Omega \rightarrow \mathbb{R}$$

$$\Rightarrow \nabla \mu(x) \cdot \nabla u(x) + \mu(x) \nabla^2 u(x) = F(x)$$

$$\vec{f} = \mu(x) \cdot \nabla u(x)$$

Heat conduction:

$$[\nabla \mu \cdot \nabla u] = \frac{\text{W}}{\text{m}^2 \text{K}} \cdot \frac{\text{K}}{\text{m}} = \frac{\text{W}}{\text{m}^3}$$

$$[\vec{f}] = [\mu(x) \cdot \nabla u(x)] = \frac{\text{W}}{\text{mK}} \cdot \frac{\text{K}}{\text{m}} = \frac{\text{W}}{\text{m}^2}$$

$$[\mu \nabla^2 u] = \frac{\text{W}}{\text{mK}} \cdot \frac{\text{K}}{\text{m}^2} = \frac{\text{W}}{\text{m}^3}$$

The physical units add up

$\Downarrow$

$$[\nabla \mu \cdot \nabla u + \mu \nabla^2 u] = [F] = \frac{\text{W}}{\text{m}^3}$$

Mass diffusion:

$$[\nabla \mu \cdot \nabla u] = \frac{\text{m}}{\text{s}} \cdot \frac{\text{mol}}{\text{m}^4} = \frac{\text{mol}}{\text{m}^3 \text{s}} = [F] = [\mu \nabla^2 u]$$

$$[\vec{f}] = \frac{\text{m}^2}{\text{s}} \cdot \frac{\text{mol}}{\text{m}^4} = \frac{\text{mol}}{\text{m}^2 \text{s}}$$

Flow in ducts

$$[\nabla \mu \cdot \nabla u] = \frac{\text{Pa} \cdot \text{s}}{\text{m}} \cdot \frac{\text{m}}{\text{s m}} = \frac{\text{Pa}}{\text{m}} = \frac{\text{N}}{\text{m}^3} = [F] = [\mu \nabla^2 u]$$

$$[\vec{f}] = \text{Pa} \cdot \text{s} \cdot \frac{\text{m}}{\text{s m}} = \text{Pa}$$

2. Find an analytical solution for the one-dimensional Poisson equation assuming pure Dirichlet boundary conditions $u(0) = u(L) = 0$, with parameters $\zeta = 0$, $\mu = 1$, and source term $f = 1$. Next, consider a piecewise source term where $f = 1$ for $x \in (0, L/2)$ and $f = 0$ otherwise. What additional interface/-continuity conditions on u are required to uniquely determine the solution?

$$\mu(x) = 1, \zeta = 1, f(x) = 1$$

$$u(L) = u(0) = 0$$

$$-\frac{d}{dx} \left(1 \frac{du}{dx} \right) = 1 \Rightarrow \frac{d^2 u}{dx^2} = 1$$

$$\Rightarrow \int \frac{d^2 u}{dx^2} dx = \int 1 dx \Rightarrow$$

$$\Rightarrow -\frac{du}{dx} = x + C \Rightarrow \int \frac{du}{dx} dx = \int x + C dx$$

$$\Rightarrow -u(x) = \frac{x^2}{2} + Cx + K$$

Using the BC:

$$u(0) = 0 \Rightarrow -\frac{0}{2} - C \cdot 0 - K = 0 \Rightarrow K = 0$$

$$u(L) = 0 \Rightarrow -\frac{L^2}{2} - CL = 0 \Rightarrow C = -\frac{L}{2}$$

$$\therefore u(x) = -\frac{1}{2}(x^2 - L)$$

Second case

$$f(x) = \begin{cases} 1, & x \in [0, L/2] \\ 0, & x \in (L/2, L] \end{cases}$$

Suppose we don't impose any continuity condition.

Then, any solution of the form

$$-u(x) = \begin{cases} \frac{x^2}{2} + C_1 x + K_1, & x \in [0, L/2] \\ C_2 x + K_2, & \text{otherwise} \end{cases}$$

With $K_1 = 0$ and $C_2 L + K_2 = 0 \Rightarrow K_2 = -C_2 L$ (because of the BCs)

is a solution to the ODE, except at $x = L/2$ (because it might not be continuous):

$$-\frac{d^2 u}{dx^2} = \begin{cases} 1, & x \in [0, L/2] = f|_{[0, L/2] \setminus \{L/2\}}(x) \\ 0, & x \in (L/2, L] \end{cases}$$

Then, we must impose that at $x = L/2$, both functions of the piecewise function u have the value. Then:

$$\frac{1}{2} \left(\frac{L}{2} \right)^2 + C_1 \frac{L}{2} = C_2 \frac{L}{2} - C_2 L \Rightarrow \frac{L}{8} + \frac{C_1}{2} = -\frac{C_2}{2} \Rightarrow C_2 = -C_1 - \frac{L}{4}$$

Still, we don't have a unique solution. We can try to look at the derivative of u :

$$-u'(x) = \begin{cases} x + C_1, & x \in [0, L/2] \\ C_2, & x \in (L/2, L] \end{cases}$$

imposing that the derivative is continuous, we must have:

$$\frac{L}{2} + C_1 = C_2$$

Now, we have a system to solve:

$$\begin{cases} C_2 - C_1 = \frac{L}{2} & (1) \\ C_2 + C_1 = -\frac{L}{4} & (2) \end{cases} \quad \begin{array}{l} \text{Substituting (1) in (2):} \\ \frac{L}{2} + C_1 + C_1 = -\frac{L}{4} \Rightarrow 2C_1 = -\frac{3L}{4} \Rightarrow C_1 = -\frac{3L}{8} \end{array}$$

$$\Rightarrow C_2 = \frac{L}{2} - \frac{3L}{8} = \frac{4L - 3L}{8} = \frac{L}{8}$$

$$\Rightarrow K_2 = -\frac{L^2}{8}$$

then,

$$u(x) = \begin{cases} -\frac{x^2}{2} + \frac{3L}{8}x, & x \in [0, L/2] \\ -\frac{1}{8}x + \frac{L^2}{8}, & x \in (L/2, L] \end{cases}$$

is the only solution such that $u \in C^1([0, L])$

To uniquely determine u , we must impose $u \in C^1([0, L])$

3. (AI) In the previous problem, assume that the source term f is a Dirac delta distribution concentrated at $x = L/2$, denoted by $\delta_{L/2}$ and defined as the generalized function satisfying

$$\int_0^L \delta_{L/2} w(x) dx = w(L/2) \quad \forall w \in C^\infty(0, L).$$

Propose or research an analytical strategy to solve this problem.

$$f(x) = \delta_{L/2} \Rightarrow \begin{cases} -\frac{d^2 u}{dx^2}(x) = \delta_{L/2} \\ u(0) = u(L) = 0 \end{cases}$$

$$\frac{du}{dx}(x) - \frac{du}{dx}(0) = \int_0^x \frac{d^2 u}{ds^2}(s) ds = -\int_0^x \delta_{L/2} ds$$

we can define

$$\delta(x) = \begin{cases} 0, & x \neq 0 \\ \infty, & x = 0 \end{cases}$$

$$\text{such that } \int_{-\infty}^{\infty} \delta(x) dx = 1$$

And

$$\delta_{L/2} = \delta(x - L/2)$$

With this definition, we have 3 cases:

Case 1: $x < L/2$

$$\int_0^x \delta_{L/2} ds = \int_0^x \delta(s - L/2) ds = \int_0^x 0 ds = 0$$

Case 2: $x > L/2$ take $\varepsilon > 0$ such that $x - L/2 > \varepsilon$

$$\int_0^x \delta_{L/2} ds = \int_0^{L/2-\varepsilon} \delta_{L/2} ds + \int_{L/2-\varepsilon}^{L/2+\varepsilon} \delta_{L/2} ds + \int_{L/2+\varepsilon}^x \delta_{L/2} ds = \int_{L/2-\varepsilon}^{L/2+\varepsilon} \delta(s - L/2) ds = 1$$

Case 3: $x = L/2$. Take $\varepsilon > 0$

$$\int_0^x \delta_{L/2} ds = \int_0^{L/2-\varepsilon} \delta_{L/2} ds + \int_{L/2-\varepsilon}^{L/2+\varepsilon} \delta_{L/2} ds$$

Knowing that $\int_{L/2-\varepsilon}^{L/2+\varepsilon} \delta_{L/2} ds = 0$

$$\int_0^x \delta_{L/2} ds = \int_{L/2-\varepsilon}^{L/2} \delta_{L/2} ds + \int_{L/2}^{L/2+\varepsilon} \delta_{L/2} ds$$

As $\varepsilon \rightarrow 0$

$$\int_0^x \delta_{L/2} ds = \int_{-\infty}^{L/2-\varepsilon} \delta_{L/2} ds + \int_{L/2-\varepsilon}^{\infty} \delta_{L/2} ds = \int_{-\infty}^{\infty} \delta_{L/2} ds = 1$$

Then

$$\frac{du}{dx}(x) = \frac{du}{dx}(0) = \begin{cases} -1, & \text{if } x > L/2 \\ 0, & \text{otherwise} \end{cases}$$

Denoting $\frac{du}{dx}(0) = C$

$$\frac{du}{dx}(x) = \begin{cases} -1+C, & \text{if } x > L/2 \\ C, & \text{otherwise} \end{cases} \Rightarrow$$

$$u(x) = \begin{cases} (-1+C)x + k_1, & \text{if } x > L/2 \\ Cx + k_2, & \text{otherwise} \end{cases}$$

imposing the BCs and u to be continuous:

$$u(L) = 0 \Rightarrow (-1+C)L + k_1 = 0 \Rightarrow k_1 = (1-C)L$$

$$u(0) = 0 \Rightarrow k_2 = 0$$

At $x = \frac{L}{2}$:

$$(-1+C)\frac{L}{2} + (1-C)L = C\frac{L}{2} \Rightarrow \frac{-1+C}{2} + 1-C = \frac{C}{2} \Rightarrow$$

$$\Rightarrow -1+C+2-2C = \frac{C}{2} \Rightarrow 1-2C = \frac{C}{2} \Rightarrow C = \frac{1}{2}$$

$$\Rightarrow k_1 = \frac{1}{2}L$$

$$\therefore u(x) = \begin{cases} -\frac{x+L}{2}, & x > L/2 \\ \frac{x}{2}, & \text{otherwise} \end{cases}$$

4. Consider Poisson's equation with $f = -6$ and $\mu = 1$ in the unit square domain $\Omega = (0, 1)^2$. Verify that

$$u(x_1, x_2) = 1 + x_1^2 + 2x_2^2 \quad (1.20)$$

is an exact solution. Which boundary conditions does this solution satisfy?

$$-\nabla^2 u(x) = -6$$

$$\frac{\partial u}{\partial x_1} = 2x_1 \quad \frac{\partial^2 u}{\partial x_1^2} = 2$$

$$\frac{\partial u}{\partial x_2} = 4x_2 \quad \frac{\partial^2 u}{\partial x_2^2} = 4$$

$$-\nabla^2 u(x) = -(2+4) = -6$$

We can determine Dirichlet and Neumann BCs in each side of the square. Then, we impose any combination of those, only remembering that there must be at least one Dirichlet BC.

Dirichlet: Applying u in the boundaries:

$$x_1 = 0; 0 \leq x_2 \leq 1$$

$$u(0, x_2) = 1 + 4x_2^2$$

$$x_1 = 1; 0 \leq x_2 \leq 1$$

$$u(1, x_2) = 2 + 4x_2^2$$

$$0 \leq x_1 \leq 1; x_2 = 0$$

$$u(x_1, 0) = 1 + x_1^2$$

$$0 \leq x_1 \leq 1; x_2 = 1$$

$$u(x_1, 1) = 5 + x_1^2$$

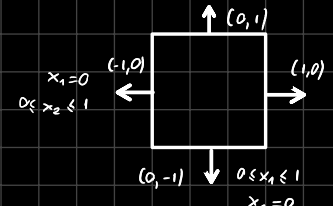
Neumann: Imposing conditions on $\nabla u(x) \cdot \vec{n}$

$$x_1 = 0; 0 \leq x_2 \leq 1$$

$$\vec{n} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$\nabla u \cdot \vec{n} = -\frac{\partial u}{\partial x_1}(0, x_2) = 0$$

$$\Rightarrow \nabla u(x_1, x_2) \cdot \vec{n} = 0$$



$$x_1 = 1; 0 \leq x_2 \leq 1$$

$$\vec{n} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\nabla u \cdot \vec{n} = \frac{\partial u}{\partial x_1}(1, x_2) = 2 \Rightarrow \nabla u(x_1, x_2) \cdot \vec{n} = 2$$

$$0 \leq x_1 \leq 1; x_2 = 0$$

$$\vec{n} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$\nabla u \cdot \vec{n} = -\frac{\partial u}{\partial x_2}(x_1, 0) = 0$$

$$\nabla u(x_1, x_2) \cdot \vec{n} = 0$$

$$0 \leq x_1 \leq 1; x_2 = 1$$

$$\vec{n} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\nabla u \cdot \vec{n} = \frac{\partial u}{\partial x_2}(x_1, 1) = 4$$

$$\nabla u(x_1, x_2) \cdot \vec{n} = 4$$

5. In two dimensions, consider a scalar function ψ (the stream function) such that

$$u_1 = -\frac{\partial \psi}{\partial x_2}, \quad u_2 = \frac{\partial \psi}{\partial x_1}. \quad (1.21)$$

Show that the resulting velocity field is solenoidal ($\nabla \cdot \mathbf{u} = 0$).

$$u: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$\nabla \cdot \mathbf{u} = \frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2} = -\frac{\partial^2 \psi}{\partial x_1 \partial x_2} + \frac{\partial^2 \psi}{\partial x_2 \partial x_1}$$

Assuming $\psi \in C^2$, using Schwarz theorem, we obtain

$$\frac{\partial^2 \psi}{\partial x_1 \partial x_2} = \frac{\partial^2 \psi}{\partial x_2 \partial x_1} \Rightarrow \nabla \cdot \mathbf{u} = 0$$

6. Using the velocity field from the previous problem, compute the corresponding Cauchy stress tensor σ .

$$\sigma(x, t) = -p(x, t)I + \mu(\nabla u + \nabla^T u)$$

$$\nabla u = \begin{bmatrix} \frac{\partial u_1}{\partial x_1} & \frac{\partial u_1}{\partial x_2} \\ \frac{\partial u_2}{\partial x_1} & \frac{\partial u_2}{\partial x_2} \end{bmatrix}$$

$$\nabla u + \nabla^T u = \begin{bmatrix} \frac{\partial u_1}{\partial x_1} + \frac{\partial u_1}{\partial x_1} & \frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} \\ \frac{\partial u_2}{\partial x_1} + \frac{\partial u_1}{\partial x_2} & \frac{\partial u_2}{\partial x_2} + \frac{\partial u_2}{\partial x_2} \end{bmatrix}$$

$$\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} = \frac{\partial^2 \psi}{\partial x_1^2} + \frac{\partial^2 \psi}{\partial x_2^2} = \frac{\partial}{\partial x_1} \left(\frac{\partial \psi}{\partial x_1} \right) - \frac{\partial}{\partial x_2} \left(\frac{\partial \psi}{\partial x_2} \right) =$$

$$= \nabla \times (\nabla \psi)$$

$$\frac{\partial u_1}{\partial x_1} = -\frac{\partial^2 \psi}{\partial x_1 \partial x_2}$$

$$\frac{\partial u_2}{\partial x_2} = \frac{\partial^2 \psi}{\partial x_2 \partial x_1}$$

$$\Rightarrow \sigma = \begin{bmatrix} -p - 2\mu \gamma_{x_1 x_2} & \nabla \times (\nabla \psi) \\ \nabla \times (\nabla \psi) & -p - 2\mu \gamma_{x_1 x_2} \end{bmatrix}$$

7. For the 3D Navier-Stokes system, write out the three scalar momentum equations in explicit component form.

Given an initial velocity field u_0 that satisfies the incompressibility constraint ($\nabla \cdot u_0 = 0$), the initial-boundary value problem for the Navier-Stokes system is stated as follows:

$$\begin{cases} \rho \left(\frac{\partial u}{\partial t} + u \cdot \nabla u \right) - \nabla \cdot (2\mu \varepsilon(u)) + \nabla p = f & \text{in } \Omega \times (0, T], \\ \nabla \cdot u = 0 & \text{in } \Omega \times (0, T], \\ u = u_D & \text{on } \Gamma_D \times (0, T], \\ [-pI + 2\mu \varepsilon(u)] \cdot n = \mathcal{F} & \text{on } \Gamma_N \times (0, T], \end{cases} \quad (1.17)$$

$$\frac{\partial u}{\partial t} = \begin{bmatrix} \frac{\partial u_1}{\partial t} \\ \frac{\partial u_2}{\partial t} \\ \frac{\partial u_3}{\partial t} \end{bmatrix}, \quad \nabla p = \begin{bmatrix} \frac{\partial p}{\partial x_1} \\ \frac{\partial p}{\partial x_2} \\ \frac{\partial p}{\partial x_3} \end{bmatrix}, \quad f = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix}$$

$$(u \cdot \nabla) u = \begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} \cdot \begin{bmatrix} \frac{\partial}{\partial x_1} \\ \frac{\partial}{\partial x_2} \\ \frac{\partial}{\partial x_3} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = u_1 \frac{\partial}{\partial x_1} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} + u_2 \frac{\partial}{\partial x_2} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} + u_3 \frac{\partial}{\partial x_3} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$$= \begin{bmatrix} u_1 \frac{\partial u_1}{\partial x_1} + u_2 \frac{\partial u_1}{\partial x_2} + u_3 \frac{\partial u_1}{\partial x_3} \\ u_1 \frac{\partial u_2}{\partial x_1} + u_2 \frac{\partial u_2}{\partial x_2} + u_3 \frac{\partial u_2}{\partial x_3} \\ u_1 \frac{\partial u_3}{\partial x_1} + u_2 \frac{\partial u_3}{\partial x_2} + u_3 \frac{\partial u_3}{\partial x_3} \end{bmatrix}$$

$$-\nabla \cdot (2\mu \varepsilon(u)) = -\mu \nabla \cdot (2\varepsilon(u)) = -\mu \nabla \cdot (\nabla u + \nabla^T u) =$$

$$= -\mu \nabla^2 u - \mu \nabla \cdot \nabla^T u = -\mu \begin{bmatrix} \nabla^2 u_1 \\ \nabla^2 u_2 \\ \nabla^2 u_3 \end{bmatrix} - \mu \begin{bmatrix} \nabla \cdot \frac{\partial u}{\partial x_1} \\ \nabla \cdot \frac{\partial u}{\partial x_2} \\ \nabla \cdot \frac{\partial u}{\partial x_3} \end{bmatrix}$$

This gives us the explicit system:

$$\rho \left(\frac{\partial u_1}{\partial t} + u_1 \frac{\partial u_1}{\partial x_1} + u_2 \frac{\partial u_1}{\partial x_2} + u_3 \frac{\partial u_1}{\partial x_3} \right) - \mu \left(\nabla^2 u_1 + \nabla \cdot \frac{\partial u}{\partial x_1} \right) + \frac{\partial p}{\partial x_1} = f_1$$

$$\rho \left(\frac{\partial u_2}{\partial t} + u_1 \frac{\partial u_2}{\partial x_1} + u_2 \frac{\partial u_2}{\partial x_2} + u_3 \frac{\partial u_2}{\partial x_3} \right) - \mu \left(\nabla^2 u_2 + \nabla \cdot \frac{\partial u}{\partial x_2} \right) + \frac{\partial p}{\partial x_2} = f_2$$

$$\rho \left(\frac{\partial u_3}{\partial t} + u_1 \frac{\partial u_3}{\partial x_1} + u_2 \frac{\partial u_3}{\partial x_2} + u_3 \frac{\partial u_3}{\partial x_3} \right) - \mu \left(\nabla^2 u_3 + \nabla \cdot \frac{\partial u}{\partial x_3} \right) + \frac{\partial p}{\partial x_3} = f_3$$

(using GPT to verify the solution, it gave this suggestion) if we consider the flow incompressible:

$$\nabla \cdot u = 0$$

$$\text{Note that } \nabla \cdot \frac{\partial u}{\partial x_j} = \nabla \cdot \left(\frac{\partial u_1}{\partial x_j}, \frac{\partial u_2}{\partial x_j}, \frac{\partial u_3}{\partial x_j} \right) =$$

$$= \frac{\partial^2 u_1}{\partial x_1 \partial x_j} + \frac{\partial^2 u_2}{\partial x_2 \partial x_j} + \frac{\partial^2 u_3}{\partial x_3 \partial x_j} = \frac{\partial}{\partial x_j} \left(\frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2} + \frac{\partial u_3}{\partial x_3} \right) = \frac{\partial}{\partial x_j} \nabla \cdot u$$

Using the incompressibility:

$$\frac{\partial}{\partial x_j} \nabla \cdot u = \frac{\partial}{\partial x_j} 0 = 0$$

Then, we simplify the system to:

$$\rho \left(\frac{\partial u_1}{\partial t} + u_1 \frac{\partial u_1}{\partial x_1} + u_2 \frac{\partial u_1}{\partial x_2} + u_3 \frac{\partial u_1}{\partial x_3} \right) - \mu \nabla^2 u_1 + \frac{\partial p}{\partial x_1} = f_1$$

$$\rho \left(\frac{\partial u_2}{\partial t} + u_1 \frac{\partial u_2}{\partial x_1} + u_2 \frac{\partial u_2}{\partial x_2} + u_3 \frac{\partial u_2}{\partial x_3} \right) - \mu \nabla^2 u_2 + \frac{\partial p}{\partial x_2} = f_2$$

$$\rho \left(\frac{\partial u_3}{\partial t} + u_1 \frac{\partial u_3}{\partial x_1} + u_2 \frac{\partial u_3}{\partial x_2} + u_3 \frac{\partial u_3}{\partial x_3} \right) - \mu \nabla^2 u_3 + \frac{\partial p}{\partial x_3} = f_3$$

8. Consider the following 2D manufactured solution for the Navier-Stokes equations:

$$u(x_1, x_2, t) = [\sin(x_1) \sin(x_2 + t), \cos(x_1) \cos(x_2 + t)]^T,$$

$$p(x_1, x_2, t) = \cos(x_1) \sin(x_2 + t).$$

Determine the body force vector f and boundary conditions corresponding to this solution.

$$\frac{\partial u}{\partial x_1} = \begin{bmatrix} \cos(x_1) \sin(x_2 + t) \\ -\sin(x_1) \cos(x_2 + t) \end{bmatrix}, \quad \frac{\partial u}{\partial x_2} = \begin{bmatrix} \sin(x_1) \cos(x_2 + t) \\ -\cos(x_1) \sin(x_2 + t) \end{bmatrix}$$

Verifying if it is incompressible

$$\nabla \cdot u = \frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2} = \cos(x_1) \sin(x_2 + t) - \cos(x_1) \sin(x_2 + t) = 0$$

$$\frac{\partial u}{\partial t} = \begin{bmatrix} \sin(x_1) \cos(x_2 + t) \\ -\cos(x_1) \sin(x_2 + t) \end{bmatrix}$$

$$\nabla^2 u_1 = -\sin(x_1) \sin(x_2 + t) - \sin(x_1) \sin(x_2 + t) = -2\sin(x_1) \sin(x_2 + t)$$

$$\nabla^2 u_2 = -\cos(x_1) \cos(x_2 + t) - \cos(x_1) \cos(x_2 + t) = -2\cos(x_1) \cos(x_2 + t)$$

$$\nabla p = \begin{bmatrix} -\sin(x_1) \sin(x_2 + t) \\ \cos(x_1) \cos(x_2 + t) \end{bmatrix}$$

Using the system from the previous exercise, simplified to 2D:

First equation:

$$\rho \left(\sin(x_1) \cos(x_2+t) + \sin(x_1) \sin(x_2+t) \cos(x_1) \sin(x_2+t) + \cos(x_1) \cos(x_2+t) \sin(x_1) \cos(x_2+t) \right) + 2\mu \sin(x_1) \sin(x_2+t) - \sin(x_1) \sin(x_2+t) = f_1$$

$$\Rightarrow f_1 = \rho \left(\sin(x_1) \cos(x_2+t) + \frac{1}{2} \sin(2x_1) \right) + (2\mu-1) \sin(x_1) \sin(x_2+t)$$

Second equation:

$$\rho \left(-\cos(x_1) \sin(x_2+t) - \sin(x_1) \sin(x_2+t) \sin(x_1) \cos(x_2+t) - \cos(x_1) \cos(x_2+t) \cos(x_1) \sin(x_2+t) \right) + 2\mu \cos(x_1) \cos(x_2+t) + \cos(x_1) \cos(x_2+t) = f_2$$

$$\Rightarrow f_2 = -\rho \left(\cos(x_1) \sin(x_2+t) + \frac{1}{2} \sin(2x_1+2t) \right) + (2\mu+1) \cos(x_1) \cos(x_2+t)$$

For BC, we can impose:

$$u(x_1, x_2) = \begin{bmatrix} \sin(x_1) \sin(x_2+t) \\ \cos(x_1) \cos(x_2+t) \end{bmatrix}, \text{ in } \Gamma_D \times (0, T]$$

$$p = \cos(x_1) \sin(x_2+t)$$

$$2\mu \varepsilon(u) = \mu (\nabla u + \nabla^T u) = \mu \left(\begin{bmatrix} \cos(x_1) \sin(x_2+t) & \sin(x_1) \cos(x_2+t) \\ -\sin(x_1) \cos(x_2+t) & -\cos(x_1) \sin(x_2+t) \end{bmatrix} + \begin{bmatrix} \cos(x_1) \sin(x_2+t) & -\sin(x_1) \cos(x_2+t) \\ \sin(x_1) \cos(x_2+t) & -\cos(x_1) \sin(x_2+t) \end{bmatrix} \right) = 2\mu \begin{bmatrix} \cos(x_1) \sin(x_2+t) & 0 \\ 0 & -\cos(x_1) \sin(x_2+t) \end{bmatrix}$$

Or

$$[-pI + 2\mu \varepsilon(u)] \cdot \vec{n} = \begin{bmatrix} (2\mu-1) \cos(x_1) \sin(x_2+t) & 0 \\ 0 & -(2\mu+1) \cos(x_1) \sin(x_2+t) \end{bmatrix} \cdot \vec{n}, \text{ in } \Gamma_N \times (0, T]$$

9. Derive the non-dimensional form of the Navier-Stokes equations in terms of the dimensionless variables $\hat{u}$, $\hat{x}$, $\hat{p}$, $\hat{t}$, and $\hat{f}$, expressing the equations in terms of the Reynolds number $Re = \frac{\rho U L}{\mu}$.

Doing the dimension analysis

$$[\rho] = \frac{kg}{m^3}, \quad \left[u_i \frac{\partial u_j}{\partial x_i} \right] = \frac{m}{s^2}$$

$$\left[\frac{\partial p}{\partial x_i} \right] = \frac{N}{m^3}$$

$$\left[\rho \left(\frac{\partial u}{\partial t} + (u \cdot \nabla) u \right) \right] = \frac{kg}{m^3} \cdot \frac{m}{s^2} = \frac{kgm}{s^2 m^3} = \frac{N}{m^3}$$

$$[\mu] = Pa \cdot s$$

$$[\varepsilon(u)] = \frac{1}{s}$$

$$[f] = \frac{N}{m^3}$$

$$\left[\nabla \cdot (2\mu \varepsilon(u)) \right] = \frac{1}{m} \cdot Pa \cdot s \cdot \frac{1}{s} = \frac{Pa}{m} = \frac{N}{m^3}$$

For the incompressibility constraint:

$$[\nabla \cdot u] = \frac{1}{s}$$

(Taken ^{partially} from the book: introduction to Fluid Dynamics, Robert W. Fox, Philip J. Pritchard, Alan T. McDonald)

$$\text{Define } \hat{x} = \frac{x}{L} \Rightarrow x = \hat{x} L; \quad \hat{u} = \frac{u}{U} \Rightarrow u = \hat{u} U$$

$$\hat{p} = \frac{p}{\rho U^2}, \quad \hat{t} = \frac{t U}{L} \Rightarrow t = \frac{\hat{t} L}{U}$$

$$\hat{f} = \frac{f L}{\rho U^2} \Rightarrow f = \frac{\hat{f} \rho U^2}{L}$$

Substituting in the equations

$$u(\hat{x}, \hat{t}) = U \hat{u}(\hat{x}, \hat{t})$$

$$\frac{\partial u}{\partial t}(\hat{x}, \hat{t}) = U \frac{\partial \hat{u}}{\partial \left(\frac{\hat{t} L}{U} \right)}(\hat{x}, \hat{t})$$

$$= U \frac{\partial \hat{u}}{\partial \hat{t}}(\hat{x}, \hat{t}) \cdot \frac{\partial \hat{t}}{\partial \left(\frac{\hat{t} L}{U} \right)} = \frac{U^2}{L} \frac{\partial \hat{u}}{\partial \hat{t}}(\hat{x}, \hat{t})$$

$$\frac{\partial u}{\partial x_i} = \frac{\partial u}{\partial (\hat{x}_i L)} = \frac{\partial u}{\partial \hat{x}_i} \cdot \frac{\partial \hat{x}_i}{\partial (\hat{x}_i L)} = \frac{1}{L} \frac{\partial u}{\partial \hat{x}_i} \Rightarrow \nabla u = \frac{1}{L} \hat{\nabla} u \Rightarrow \hat{\nabla} = L \nabla$$

$$\Rightarrow (u \cdot \nabla) u = U^2 \left(\hat{u} \cdot \frac{1}{L} \hat{\nabla} \right) \hat{u} = \frac{U^2}{L} (\hat{u} \cdot \hat{\nabla}) \hat{u}$$

With this, we also have:

$$\nabla \cdot u = \frac{1}{L} \hat{\nabla} \cdot u \Rightarrow$$

$$\nabla \cdot (2\mu \varepsilon(u)) = \frac{1}{L} \hat{\nabla} \cdot \left(\frac{U \mu}{L} (\hat{\nabla} \hat{u} + \hat{\nabla}^T \hat{u}) \right) = \frac{U \mu}{L^2} \hat{\nabla} \cdot (\hat{\nabla} \hat{u} + \hat{\nabla}^T \hat{u})$$

$$\nabla p = \frac{1}{L} \hat{\nabla} (U \hat{p}) = \frac{U}{L} \hat{\nabla} \hat{p}$$

$$\rho \left(\frac{\partial u}{\partial t} + (u \cdot \nabla) u \right) - \nabla \cdot (2\mu \varepsilon(u)) + \nabla p = f \Rightarrow$$

$$\Rightarrow \rho \left(\frac{U^2}{L} \frac{\partial \hat{u}}{\partial \hat{t}} + \frac{U^2}{L} (\hat{u} \cdot \hat{\nabla}) \hat{u} \right) - \frac{U \mu}{L^2} \hat{\nabla} \cdot (\hat{\nabla} \hat{u} + \hat{\nabla}^T \hat{u}) + \frac{U \hat{p}}{L} \hat{\nabla} \hat{p} = \frac{\rho U^2}{L} \hat{f}$$

$$= \rho \frac{U^2}{L} \hat{f} \Rightarrow$$

$$\Rightarrow \rho \frac{U^2}{L} \left(\frac{\partial \hat{u}}{\partial \hat{t}} + (\hat{u} \cdot \hat{\nabla}) \hat{u} \right) - \frac{\mu}{L^2} \hat{\nabla} \cdot (\hat{\nabla} \hat{u} + \hat{\nabla}^T \hat{u}) + \frac{U \hat{p}}{L} \hat{\nabla} \hat{p} = \rho \frac{U^2}{L} \hat{f}$$

Multiplying everything by $\frac{L}{U^2 \rho}$

$$\frac{\partial \hat{u}}{\partial \hat{t}} + (\hat{u} \cdot \hat{\nabla}) \hat{u} - \frac{\mu}{L U \rho} \hat{\nabla} \cdot (\hat{\nabla} \hat{u} + \hat{\nabla}^T \hat{u}) + \hat{p} \hat{\nabla} \hat{p} = \hat{f} \Rightarrow$$

$$\Rightarrow \frac{\partial \hat{u}}{\partial \hat{t}} + (\hat{u} \cdot \hat{\nabla}) \hat{u} - \frac{1}{Re} \hat{\nabla} \cdot (\hat{\nabla} \hat{u} + \hat{\nabla}^T \hat{u}) + \hat{p} \hat{\nabla} \hat{p} = \hat{f}$$

For the incompressibility constraint:

$$\nabla \cdot u = 0 \Rightarrow \frac{U}{L} \hat{\nabla} \cdot \hat{u} = 0 \Rightarrow \hat{\nabla} \cdot \hat{u} = 0$$

10. Streamlines and pathlines

- Show that Eq. (1.18) is equivalent to Eq. (1.19).
- Consider the unsteady flow

$$u = u_0, \quad v = kt, \quad w = 0,$$

where u_0 and k are positive constants.

- Show that at each fixed instant t the streamlines (cf. Definition 1.3.1) are straight lines, and determine their slope.
- Show that any fluid particle follows a parabolic path (cf. Definition 1.3.2) as time proceeds.
- Explain why the two families of curves do not coincide, and identify the single point at which a given streamline is tangent to a given pathline.

►

Definition 1.3.1 (Streamline) Let t be held fixed. A streamline through a point $x_0 \in \Omega$ is a parametric curve $\gamma(s) = (x(s), y(s), z(s))$ with $\gamma(s_0) = x_0$, whose tangent is everywhere parallel to the velocity field evaluated at that instant, i.e.

$$\gamma'(s) \times \mathbf{u}(\gamma(s), t) = \mathbf{0} \quad \text{for all } s. \quad (1.18)$$

Equivalently, writing $\mathbf{u} = (u, v, w)$,

$$\frac{dx/ds}{u} = \frac{dy/ds}{v} = \frac{dz/ds}{w}, \quad \text{with } t \text{ held fixed,} \quad (1.19)$$

where a vanishing denominator is understood to force the corresponding numerator to vanish as well.

$$1.18 \Leftrightarrow 1.19$$

$$\gamma'(s) \times \mathbf{u}(\gamma(s), t) = \begin{vmatrix} i & j & k \\ x' & y' & z' \\ u & v & w \end{vmatrix} = (y'w - z'v)i - (x'w - z'u)j + (x'v - y'u)k = \mathbf{0}$$

$$\Leftrightarrow \begin{cases} y'w - z'v = 0 \\ x'w - z'u = 0 \\ x'v - y'u = 0 \end{cases} \Leftrightarrow \begin{cases} y'w = z'v \\ x'w = z'u \\ x'v = y'u \end{cases} \Leftrightarrow \begin{cases} \frac{1}{v} \frac{dy}{ds} = \frac{1}{w} \frac{dz}{ds} \\ \frac{1}{u} \frac{dx}{ds} = \frac{1}{w} \frac{dz}{ds} \\ \frac{1}{u} \frac{dx}{ds} = \frac{1}{v} \frac{dy}{ds} \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \frac{1}{u} \frac{dx}{ds} = \frac{1}{v} \frac{dy}{ds} = \frac{1}{w} \frac{dz}{ds}$$

$$\begin{aligned} D \quad u &= u_0 \\ v &= kt \\ w &= 0 \end{aligned}$$

a) Fix $t=t_0$ and x_0 such that $\gamma(s_0)=x_0$.

As γ' is parallel the velocity field:

$$\begin{aligned} x'(s) &= u_0 & x(s) &= u_0 s + C_1 \\ y'(s) &= kt_0 & y(s) &= kt_0 s + C_2 \\ z'(s) &= 0 & z(s) &= C_3 \end{aligned}$$

We can see that γ is a line with direction vector

$$\begin{bmatrix} u_0 \\ kt_0 \\ 0 \end{bmatrix}$$

b)

Definition 1.3.2 (Pathline) The pathline of the fluid particle located at x_0 at time t_0 is the trajectory it traces out as time proceeds. It is the solution of the initial value problem (IVP)

$$\frac{dx}{dt} = \mathbf{u}(\mathbf{x}(t), t), \quad \mathbf{x}(t_0) = \mathbf{x}_0,$$

that is, in components,

$$\frac{dx}{dt} = u(\mathbf{x}(t), t), \quad \frac{dy}{dt} = v(\mathbf{x}(t), t), \quad \frac{dz}{dt} = w(\mathbf{x}(t), t).$$

$$\frac{d\vec{x}}{dt} = \vec{u}(\vec{x}(t), t), \quad \vec{x}(t_0) = \vec{x}_0$$

$$x'(t) = u(x(t), t) = u_0 \Rightarrow x(t) = u_0 t + C_1$$

$$y'(t) = v(y(t), t) = kt \Rightarrow y(t) = \frac{k}{2} t^2 + C_2$$

$$z'(t) = w(z(t), t) = 0 \Rightarrow z(t) = C_3$$

$$x(t_0) = x_0 \Rightarrow u_0 t_0 + C_1 = x_0 \Rightarrow C_1 = x_0 - u_0 t_0$$

$$y(t_0) = y_0 \Rightarrow \frac{k}{2} t_0^2 + C_2 = y_0 \Rightarrow C_2 = y_0 - \frac{k}{2} t_0^2$$

$$C_3 = z_0$$

$$\Rightarrow \vec{x}(t) = \left(x_0 + u_0(t-t_0), y_0 + \frac{k}{2}(t^2 - t_0^2), z_0 \right)$$

Looking at the plane $z=z_0$ we can write $y(x(t))$

$$x(t) = x_0 + u_0(t-t_0) \Rightarrow (x-x_0) \cdot \frac{1}{u_0} + t_0 = t \Rightarrow$$

$$\Rightarrow y(x) = y_0 + \frac{k}{2}(t-t_0)(t+t_0) = y_0 + \frac{k}{2} \left(\frac{x-x_0}{u_0} + t_0 - t_0 \right)$$

$$\left(\frac{x-x_0}{u_0} + t_0 + t_0 \right) = y_0 + \frac{k}{2} \left(\left(\frac{x-x_0}{u_0} \right)^2 + \frac{2t_0}{u_0} (x-x_0) \right) =$$

which represents a parabola

c) The two families of curves do not coincide because the velocity field changes as a function of time. The streamline is the tangent vector to the velocity field at a specific point in time. Supposing that a particle can move with the time stopped, it would follow the path of the streamline.

The pathline is the path a particle takes.

Fix $t=t_0$. Define γ the stream line that passes through $\vec{x}_0$. $\gamma(s) = (u_0 s + C_1, kt_0 s + C_2, C_3)$ with tangent vector $\vec{s} = (u_0, kt_0, 0)$

The tangent vector to a pathline is $\vec{u} = (u_0, kt_0, 0)$

Then, fixed $t=t_0$, for every particle, the streamline that passes through it is parallel to its pathline at the particle's position.